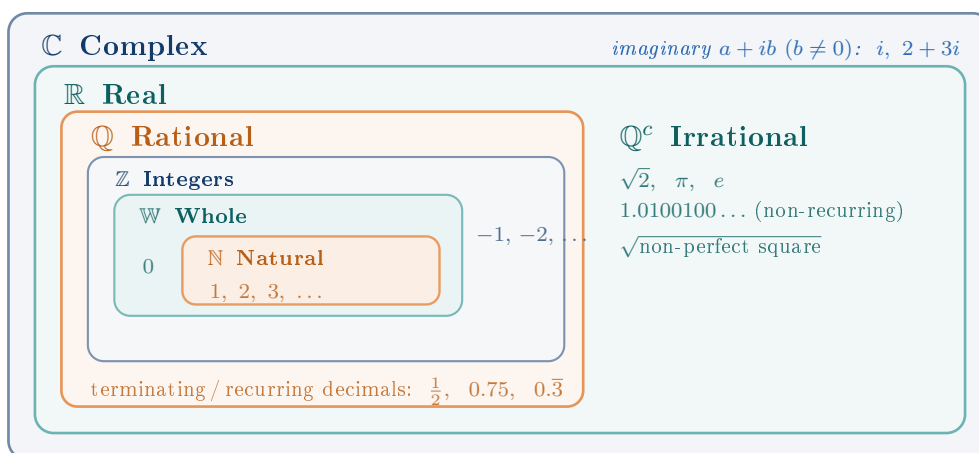


1 Number System

Every number met in JEE mathematics falls into one of the following families.

- **Natural** $\mathbb{N} = \{1, 2, 3, \dots\}$; **Whole** $\mathbb{W} = \{0, 1, 2, \dots\}$; **Integers** $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$.
- **Rational** $\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$ — decimals that *terminate* or *recur*.
- **Irrational** $\mathbb{Q}^c$ — non-terminating, non-recurring (e.g. $\sqrt{2}$, π , e).
- **Real** $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$; **Complex** $\mathbb{C} = \{a + ib : a, b \in \mathbb{R}\}$, with $i^2 = -1$.
- **Even** $= 2n$, **odd** $= 2n \pm 1$; **prime** (exactly two divisors), **composite**, **co-prime** ($\gcd = 1$).
- **Chain:** $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.



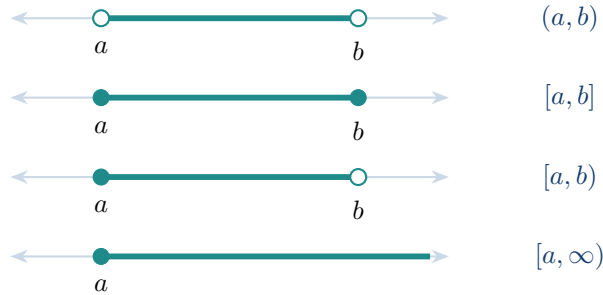
Each set sits inside the next: $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. Rationals and irrationals together make up all real numbers.

Remember: 0 is neither positive nor negative; 1 is neither prime nor composite; 2 is the only even prime. Every integer is rational; $\sqrt{\text{non-perfect square}}$ is irrational; and the sum/product of a non-zero rational with an irrational is always irrational.

2 Intervals on the Real Line

A continuous set of real numbers is written as an *interval*.

Interval	Set-builder form
(a, b)	$\{x : a < x < b\}$ (open — both ends excluded)
$[a, b]$	$\{x : a \leq x \leq b\}$ (closed — both ends included)
$[a, b), (a, b]$	$\{x : a \leq x < b\}, \{x : a < x \leq b\}$ (half-open)
$[a, \infty), (-\infty, b)$	$\{x : x \geq a\}, \{x : x < b\}$
$(-\infty, \infty)$	$\mathbb{R}$



Filled dot = endpoint included, open dot = endpoint excluded. Length of $(a, b) = b - a$.

3 Basic Algebraic Identities

- $(a \pm b)^2 = a^2 \pm 2ab + b^2$; $(a + b)(a - b) = a^2 - b^2$.
- $(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3 = a^3 \pm b^3 \pm 3ab(a \pm b)$.
- $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$.
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$.
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.
- If $a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$.
- $a^2 + b^2 = (a + b)^2 - 2ab$; $(a + b)^2 - (a - b)^2 = 4ab$.

Example. If $a + \frac{1}{a} = 3$, then $a^2 + \frac{1}{a^2} = \left(a + \frac{1}{a}\right)^2 - 2 = 9 - 2 = 7$ and $a^3 + \frac{1}{a^3} = \left(a + \frac{1}{a}\right)^3 - 3\left(a + \frac{1}{a}\right) = 27 - 9 = 18$.

4 Indices & Surds (Laws of Exponents)

For $a, b > 0$ and real m, n :

- $a^m \cdot a^n = a^{m+n}$; $\frac{a^m}{a^n} = a^{m-n}$; $(a^m)^n = a^{mn}$.
- $(ab)^n = a^n b^n$; $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$.
- $a^0 = 1$ ($a \neq 0$); $a^{-n} = \frac{1}{a^n}$; $a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$.
- **Surds:** $\sqrt{a} \sqrt{b} = \sqrt{ab}$; rationalise: $\frac{1}{\sqrt{a} + \sqrt{b}} = \frac{\sqrt{a} - \sqrt{b}}{a - b}$.
- $a^x = a^y \Rightarrow x = y$ ($a > 0, a \neq 1$); $a^x = b^x \Rightarrow a = b$ ($x \neq 0$).

Example. $\frac{2^{n+1} \cdot 4^n}{8^{n-1}} = \frac{2^{n+1} \cdot 2^{2n}}{2^{3n-3}} = 2^{(n+1)+2n-(3n-3)} = 2^4 = 16$.

5 Logarithms

Definition: $\log_a N = x \iff a^x = N$, valid for $a > 0, a \neq 1, N > 0$.

- $\log_a(mn) = \log_a m + \log_a n$; $\log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$.
- $\log_a(m^n) = n \log_a m$; $\log_{a^n} m = \frac{1}{n} \log_a m$; $\log_{a^k} m^p = \frac{p}{k} \log_a m$.
- **Change of base:** $\log_a N = \frac{\log_b N}{\log_b a}$; $\log_a b \cdot \log_b a = 1$, i.e. $\log_a b = \frac{1}{\log_b a}$.
- $\log_a a = 1$; $\log_a 1 = 0$; $a^{\log_a N} = N$; $a^{\log_c b} = b^{\log_c a}$.

- Common log $\log_{10}$; natural log $\ln = \log_e$ where $e \approx 2.718$.

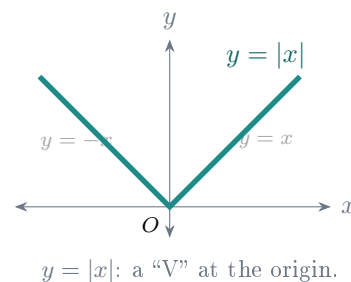
Example. $\log_2(\log_3 81) = \log_2(\log_3 3^4) = \log_2 4 = 2$. Also $2^{\log_2 7} = 7$, $\log_5 \frac{1}{25} = -2$, $\log_9 27 = \frac{3}{2}$.

Domain first: a logarithm exists only when its *argument is positive* and the *base is* $> 0, \neq 1$. Check this before applying any log law — otherwise you may introduce or lose roots.

6 Modulus (Absolute Value)

Definition: $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$ — the distance of x from 0.

- $|x| \geq 0$; $\sqrt{x^2} = |x|$; $|xy| = |x||y|$; $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$.
- Triangle:** $|x + y| \leq |x| + |y|$ and $||x| - |y|| \leq |x - y|$.
- $|x| = a \iff x = \pm a$ ($a \geq 0$).
- $|x| < a \iff -a < x < a$; $|x| > a \iff x < -a$ or $x > a$.



Example. Solve $|x - 2| < 3$: $-3 < x - 2 < 3 \Rightarrow -1 < x < 5$, i.e. $x \in (-1, 5)$.

7 Ratio, Proportion & Variation

- Ratio** $a : b = \frac{a}{b}$; **proportion** $a : b = c : d \iff ad = bc$ ($b, d \neq 0$).
- Invertendo** $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{b}{a} = \frac{d}{c}$; **Alternendo** $\frac{a}{c} = \frac{b}{d}$.
- Componendo** $\frac{a+b}{b} = \frac{c+d}{d}$; **Dividendo** $\frac{a-b}{b} = \frac{c-d}{d}$.
- Componendo-dividendo:** $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.
- If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$ then each ratio $= \frac{a+c+e}{b+d+f}$.
- Direct variation** $y \propto x \Rightarrow y = kx$; **inverse** $y \propto \frac{1}{x} \Rightarrow xy = k$.

Example. If $\frac{a}{b} = \frac{3}{5}$, then by componendo-dividendo $\frac{a+b}{a-b} = \frac{3+5}{3-5} = \frac{8}{-2} = -4$.

8 Polynomials, Remainder & Factor Theorem

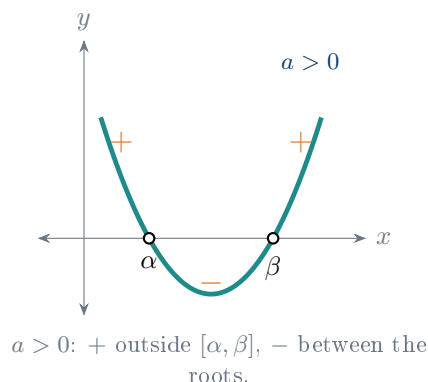
- A **polynomial** of degree n : $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$, $a_n \neq 0$.
- Division algorithm:** $p(x) = (x - a)q(x) + r$, where r is a constant.
- Remainder theorem:** the remainder when $p(x)$ is divided by $(x - a)$ is $p(a)$. For $(ax - b)$ it is $p\left(\frac{b}{a}\right)$.
- Factor theorem:** $(x - a)$ is a factor of $p(x) \iff p(a) = 0$.
- A polynomial of degree n has exactly n roots (counting multiplicity).

Example. $p(x) = x^3 - 3x + 2$: since $p(1) = 1 - 3 + 2 = 0$, $(x-1)$ is a factor. Dividing, $p(x) = (x-1)(x^2 + x - 2) = (x-1)^2(x+2)$.

9 Quadratic Equations — Basics

For $ax^2 + bx + c = 0$ ($a \neq 0$): $x = \frac{-b \pm \sqrt{D}}{2a}$, where $D = b^2 - 4ac$ (discriminant).

- **Nature of roots:** $D > 0$ real & distinct; $D = 0$ real & equal ($x = -\frac{b}{2a}$); $D < 0$ complex conjugates.
- **Sum** $\alpha + \beta = -\frac{b}{a}$; **product** $\alpha\beta = \frac{c}{a}$.
- **Equation from roots:** $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.
- **Sign of $ax^2 + bx + c$** (when $a > 0$, $D > 0$): positive *outside* the roots, negative *between* them.

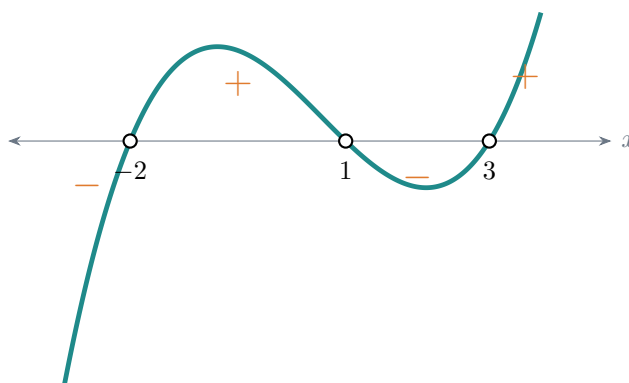


Example. $x^2 - 5x + 6 = 0$: $D = 25 - 24 = 1 > 0$ (real, distinct). Roots $x = \frac{5 \pm 1}{2} = 2, 3$; sum $= 2 + 3 = 5 = -\frac{b}{a} = -\frac{-5}{1}$, product $= 6 = \frac{c}{a} = \frac{6}{1}$. ✓

10 Inequalities & the Wavy-Curve Method

- Adding/subtracting the same number keeps an inequality; **multiplying or dividing by a negative reverses it**.
- **Wavy-curve (method of intervals):** write the expression as a product/quotient of *linear* factors, mark the roots on a number line, and draw a curve that starts + from the far right and *alternates sign* at each simple root. At a factor of *even* multiplicity the sign does *not* change.
- **Modulus:** $|x| < a \iff -a < x < a$; $|x| > a \iff x < -a$ or $x > a$ ($a > 0$).

sign of $(x+2)(x-1)(x-3)$



Sign scheme of $(x+2)(x-1)(x-3)$: the sign alternates across the simple roots $-2, 1, 3$.

Example. $(x+2)(x-1)(x-3) > 0$: from the scheme the expression is positive on $(-2, 1) \cup (3, \infty)$.

Fundamental Inequality ($AM \geq GM \geq HM$)

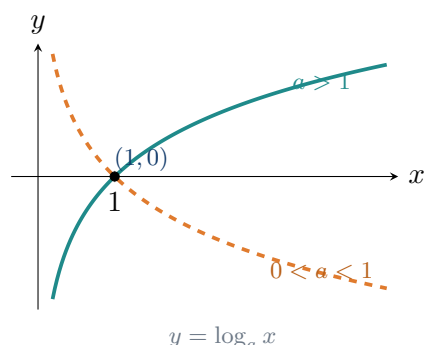
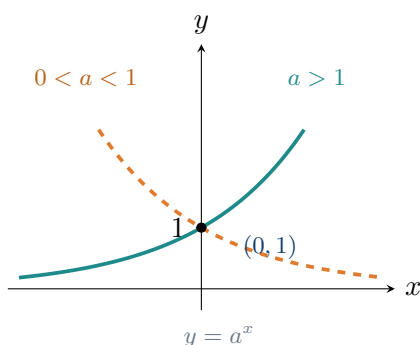
- For positive reals a, b : $\frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a+b}$, with equality iff $a = b$.

- In general, $\frac{a_1 + a_2 + \dots + a_n}{n} \geq (a_1 a_2 \dots a_n)^{1/n}$ (AM $\geq$ GM).
- **Consequences:** $a + \frac{1}{a} \geq 2$ for $a > 0$; if $a + b$ is fixed then ab is maximum when $a = b$.

11 Exponential & Logarithmic Inequalities

The direction depends entirely on whether the base is > 1 or between 0 and 1.

- $a^{f(x)} > a^{g(x)}$: if $a > 1$ then $f(x) > g(x)$; if $0 < a < 1$ then $f(x) < g(x)$ (reverse).
- $\log_a f(x) > \log_a g(x)$: if $a > 1$ then $f > g > 0$; if $0 < a < 1$ then $0 < f < g$.
- $\log_a x > 0$: if $a > 1$ then $x > 1$; if $0 < a < 1$ then $0 < x < 1$.
- a^x is increasing for $a > 1$ and decreasing for $0 < a < 1$, always through $(0, 1)$; $\log_a x$ passes through $(1, 0)$.



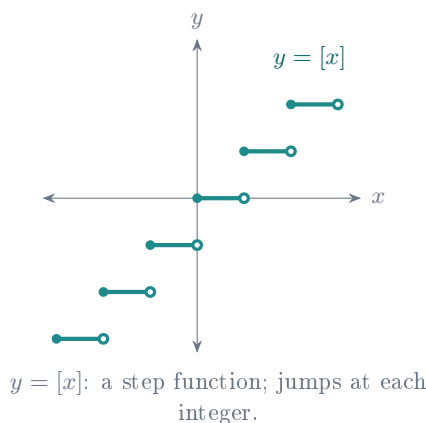
Base > 1 (solid) rises; base between 0 and 1 (dashed) falls — this fixes every inequality's direction.

Example. $\left(\frac{1}{2}\right)^x > \frac{1}{8}$: base < 1 , so *reverse* $\Rightarrow x < 3$. $\log_2 x < 3 \Rightarrow 0 < x < 2^3 = 8$.

12 Greatest Integer & Fractional Part

Greatest integer (floor) $[x]$ = greatest integer $\leq x$; **fractional part** $\{x\} = x - [x]$.

- $[x] \leq x < [x] + 1$; $0 \leq \{x\} < 1$.
- $[x + n] = [x] + n$ and $\{x + n\} = \{x\}$ for any integer n .
- $[x] = n \iff n \leq x < n + 1$; e.g. $[3.7] = 3$, $[-1.2] = -2$, $\{-1.2\} = 0.8$.
- $[x] + [-x] = \begin{cases} 0, & x \in \mathbb{Z} \\ -1, & x \notin \mathbb{Z} \end{cases}; \quad x = [x] + \{x\}.$



Example. $[2.9] + \{2.9\} = 2 + 0.9 = 2.9$; and $[\sqrt{2}] = 1$ since $1 \leq \sqrt{2} < 2$, so $\{\sqrt{2}\} = \sqrt{2} - 1$.

13 Quick Recall & Common Mistakes

Remember

- $\sqrt{x^2} = |x|$ (not x).
- A log needs argument > 0 and base $> 0, \neq 1$.
- Multiplying an inequality by a negative *flips* it.
- $a^0 = 1$ only for $a \neq 0$; 0^0 is indeterminate.
- $D = b^2 - 4ac$ decides the nature of the roots.
- $|x| > a$ holds for *all* x when $a < 0$; $|x| < a$ has no solution when $a \leq 0$.

Avoid these slips

- $\log(m + n) \neq \log m + \log n$.
- $(a + b)^2 \neq a^2 + b^2$.
- $|a + b| \neq |a| + |b|$ in general (only $\leq$).
- $x^2 = 4 \Rightarrow x = \pm 2$ — don't drop -2 .
- Never cancel a factor across an inequality — use the sign scheme.
- $\sqrt{a}\sqrt{b} = \sqrt{ab}$ needs $a, b \geq 0$; and $\log a^2 = 2 \log |a|$.

Worth memorizing: $\sqrt{2} \approx 1.41$, $\sqrt{3} \approx 1.73$, $e \approx 2.72$, $\pi \approx 3.14$; $\log_{10} 2 \approx 0.301$, $\log_{10} 3 \approx 0.477$; $\ln 2 \approx 0.693$. A quadratic has real roots iff $b^2 \geq 4ac$, and (with rational coefficients) rational roots iff D is a perfect square. $a^2 + b^2 \geq 2ab$ and $a^2 + b^2 + c^2 \geq ab + bc + ca$ for all reals.